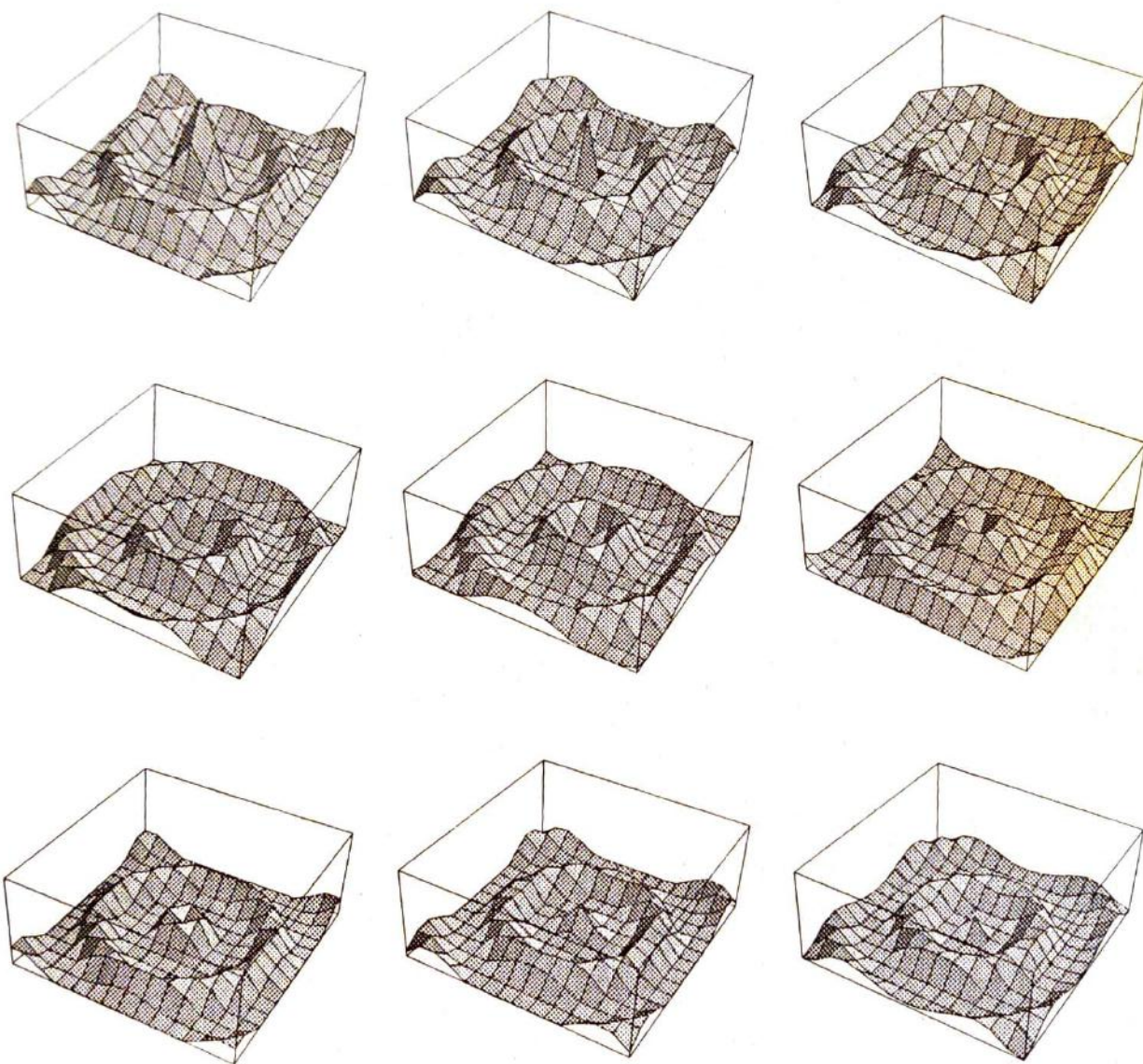


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From the Editors-in-Chief

June 1996

After battling term papers, class tests, and AP exams, we bring to you, the reader, issue number two of this year's *Math Survey*. This is the seventieth volume of *Math Survey*, and as in past volumes it is full of engaging articles and entertaining recreational mathematics.

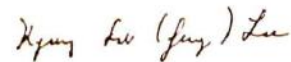
Math Survey greatly appreciates the counsel and support of Mr. Clancy, *Math Survey*'s faculty advisor, and of Dr. Rothenburg and the Stuyvesant High School mathematics department. We thank the Parents' Association, the Alumni Association, and *Math Survey*'s advertisers and sponsors for their financial support. Finally, we would like to thank the staff and editors for the time and energy they invested in this issue of *Math Survey*.

Sincerely,

Hyun-Sup Byun



Kyung Sub (Jerry) Lee



An Analog of Pick's Theorem for Circles

BY BOBBY WOODS-CORWIN

The area of a closed rectilinear graph with vertices on lattice points is known as a function of the number of lattice points contained within the graph and the number of lattice points on the graph. This relationship is expressed by Pick's Theorem, where the area of the graph is one less than the number of interior lattice points added to half the number of lattice points on the boundary of the graph, or, in symbols: $A = I + B/2 - 1$.

However, the relationship between area and lattice points for a similar curvilinear graph is more complex. For circular graphs of radius r , the area can be related to the number of interior lattice points and to the number of boundary lattice points. The area is clearly πr^2 .

The number of boundary points on a circle can be expressed as the number of ways that the radius squared can be expressed as a sum of two squares. This is found using modulo arguments and prime factorization. The number of interior lattice points can be found by an organized counting method, or by an application of Pick's Theorem to a polygon inscribed in the circle. The relationships are all found exactly for all real values of the radius of the circle.

To find the relationship between the number of lattice points in and on a circle and the radius of the circle, we examine a circle with the center at a lattice point. By symmetry, the problem can be reduced to a single quadrant, because each quadrant clearly has the same area, number of interior lattice points, and number of boundary lattice points since the circle is symmetric about the axes.

The area, $A(r)$, of a circle is simply πr^2 . Finding the number of boundary lattice points is more complex. Define $B(r)$ to be the number of lattice points on the boundary of a circle of radius r . $B(r)$ is equivalent to the number of ways that r^2 can be represented as the sum of two squares. If r^2 is not an integer, $B(r) = 0$, because $x^2 + y^2 = r^2$ has no solution in integers.

If r^2 is an integer and r is not an integer, $B(r) \equiv 0 \pmod{8}$ by symmetry, because for each (x,y) such that $x^2 + y^2 = r^2$, there are 8 lattice points, namely $(\pm x, \pm y)$ and $(\pm y, \pm x)$. Similarly, $B(r) \equiv 4 \pmod{8}$ in the case where r is an integer, because there are four additional points at $(\pm r, 0)$ and $(0, \pm r)$.

Lemma: The prime factors in r which are 2 or 3 mod 4 have no effect on $B(r)$.

Proof: If r is even, $r^2 = x^2 + y^2 \equiv 0 \pmod{4}$. Since x^2 and $y^2 \equiv 0$ or $1 \pmod{4}$ and $x^2 + y^2 \equiv 0 \pmod{4}$, x^2 and $y^2 \equiv 0 \pmod{4}$, and x and y are even. Therefore, the Pythagorean triple (x,y,r) can be reduced whenever r is even to $(x/2, y/2, r/2)$. This means that there is a one to one correspondence between $B(r)$ and $B(r/2)$ for all even r . Repeating this reduction until r is odd shows that any power of 2 as a prime factor of r has no effect on $B(r)$.

All primitive Pythagorean triples are of the form $(m^2 - n^2, 2mn, m^2 + n^2)$, where m, n are positive integers. r must be a sum of two squares, and therefore $r \equiv 0, 1$, or $2 \pmod{4}$. Therefore, there are no primitive Pythagorean triples with $r \equiv 3 \pmod{4}$. Since there are no primitive triples with $r \equiv 3 \pmod{4}$, then all powers of all prime factors $q \equiv 3 \pmod{4}$ have no effect on $B(r)$ because $B(r/q) = B(r)$. This reduction can similarly be continued until r contains only factors which are congruent to 1 mod 4.

Therefore, the only prime factors of r that affect on $B(r)$ are congruent to 1 mod 4.

Define $b(r)$ to be the number of unordered nondegenerate Pythagorean triples with r as the hypotenuse. Given that r is only affected by prime factors that are $1 \pmod{4}$, a recursive argument can be used to find $b(r)$. For each $n \equiv 1 \pmod{4}$, there is exactly one primitive Pythagorean triple with n as the hypotenuse. r , a multiple of n , will be the hypotenuse of one Pythagorean triangle which can be reduced to the primitive triple with n as hypotenuse. If $m \equiv 1 \pmod{4}$ and is not a factor of n , $b(mn) = 2b(n)$. This follows because for each Pythagorean triple counted in $b(n)$ from a specific factor k of n , there is another triple created from mk .

If m is $1 \pmod{4}$ and m^x is a factor of n , $b(mn) = b(n) + b(n/m^x)$, or equivalently, by repeating the process x times, $b(mn) = (x + 2)b(n)$. This similarly follows because only the triples corresponding to factors k of n not containing m as a prime factor (factors k of n/m^x) create a new triple from mk . If a triple was created from a factor of n containing m as a prime factor, it would just be a duplicate of an already existing triple and would thereby not add to $b(mn)$.

Extending the above recursive relation to a general case, the value of $b(r)$ can be computed. If the prime factors of r that are $1 \pmod{4}$ are labeled p_j , and they are raised to the α_j th power respectively, and the prime factors of r that are $3 \pmod{4}$ are labeled q_j , and they are raised to the β_j th power respectively, the prime factorization of r is

$$2^x \times \prod_j (p_j^{\alpha_j}) \times \prod_j (q_j^{\beta_j})$$

By applying the recursive relation above for all prime factors m , the recursive formula for finding $b(r)$ can be written in product notation as:

$$b(r) = \prod_{j \in I(4)} (1 + \alpha_j)$$

$B(r) = 8b(r) - 4$, because of symmetry and the need to add the degenerate Pythagorean triples, so,

$$B(r) = 8 \left(\prod_{j \in I(4)} (1 + \alpha_j) \right) - 4$$

Define $I(r)$ to be the number of lattice points inside a circle of radius r , not including those on the boundary. If the center were not at a lattice point, $I(r)$ could vary by a small amount, but for simplicity, we assume that the center of the circle is at a lattice point. If the lattice point is not the origin, we can move the center to the origin by translation without affecting anything else, so the circle may be assumed to be centered at the origin.

By line symmetry with the x and y axes, the four quadrants each have the same number of interior lattice points. We define that number to be $i(r)$. $4 \cdot i(r)$ plus the number of lattice points on the axes inside the circle is $I(r)$. The number of interior lattice points on the axes is $4[r]^* - 3$, because there is a lattice point on each axis from 1 to $[r] - 1$, and there is a point at the origin. The remaining points are contained within the four quadrants. They can be counted by the brute force method of just scanning across each x value and counting the number of lattice points on it. This number of lattice points for each x value is $\left[\sqrt{r^2 - x^2} \right]^\dagger$, the largest y value such that $x^2 + y^2 < r^2$.

Therefore:

$^\dagger [r]$ represents the least integer greater than or equal to r .

$^* [r]$ represents the greatest integer less than or equal to r .

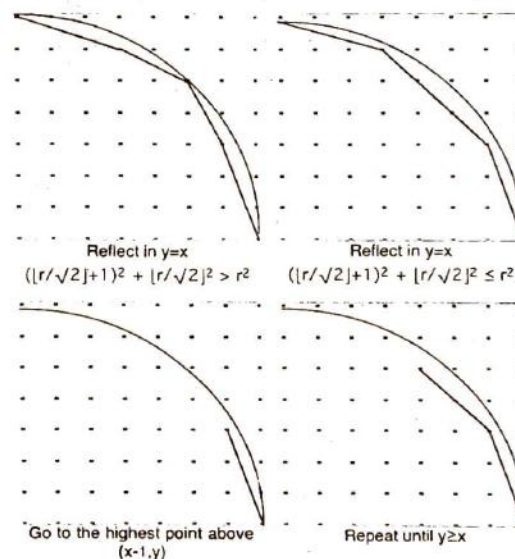
$$I(r) = 4 \left(\sum_{x=1}^{\lceil r \rceil - 1} \left\lfloor \sqrt{r^2 - x^2} \right\rfloor \right) + 4\lceil r \rceil - 3$$

However, this expression of the number of interior lattice point uses the greatest integer function. Since the expression was derived by counting, it is inherently not continuous. A different way to find the number of interior lattice points would be helpful.

A polygon containing all interior lattice points is constructed inside the circle. Using Pick's Theorem, the number of interior lattice points can be calculated from the number of boundary lattice points and the area. The number of boundary points can be found by using symmetry. The following algorithm creates 1/8 of a polygon containing all of the circle's interior lattice points.

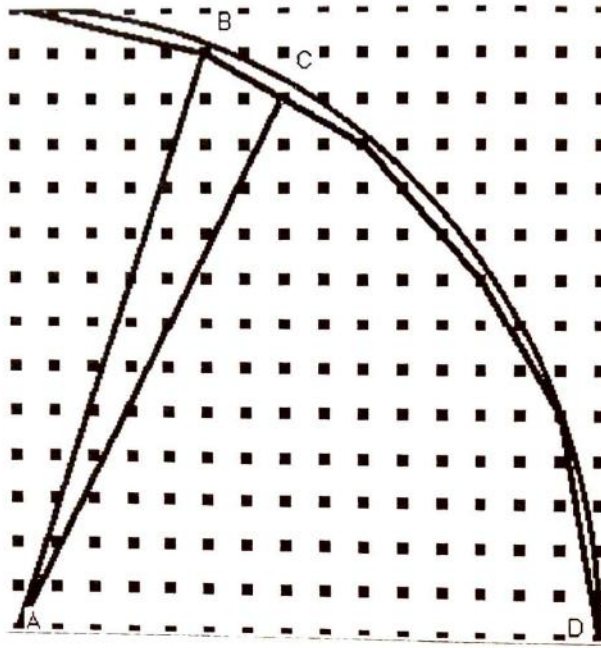
1. Start at $(r,0)$.
2. Call the present point (x,y) .
3. Go to the point $(x-1,y)$.
4. Go up until the next point is outside the circle.
5. Stop one point before leaving the circle.
6. If $x > y$ then go to step 2, else end.

Creating the interior polygon



If the resulting set of segments is reflected in the line $y=x$, the result will be a full quadrant of segments. If this new set of segments is reflected in the x and y axes, a full polygon containing all of the circle's interior lattice points is created. Each segment has two boundary points, and each point has two segments, so the number of segments is equal to the number of points. Therefore, the total number of boundary points on the polygon is 8 times $\lceil r \rceil + \lceil r/\sqrt{2} \rceil + 1$, the number of segments created by the algorithm. However, in some cases (like from $(3,4)$ to $(4,3)$) a segment is duplicated by the first reflection, so if $(\lceil r/\sqrt{2} \rceil + 1)^2 + (\lceil r/\sqrt{2} \rceil)^2 \leq r^2$ then the total number of boundary points on the polygon is $8(\lceil r \rceil + \lceil r/\sqrt{2} \rceil + 1) - 4$. The procedures described in this section are illustrated at the end.

The area of the polygon is computed by breaking the polygon into triangles. Each triangle has a vertex at the center of the circle and two consecutive vertices on the polygon. The area of one of these triangles is described in terms of the coordinates of its vertices. The center of the circle, $(0,0)$, is point A, the first polygon vertex, (x_1, y_1) , is point B, and the second polygon vertex, (x_2, y_2) , is point C, as in the figure below. Point D is $(r, 0)$.



Define r_1 as the distance AB, and r_2 as the distance AC. Define β_1 as the angle, and β_2 as the angle CAD. Define angle α as $\beta_1 - \beta_2$, or BAC.

$$\begin{aligned}\text{Area} &= ((AB)(AC) \sin(\alpha))/2 \\ &= ((r_1 r_2) \sin(\beta_1 - \beta_2))/2 \\ &= (r_1 r_2)(\sin \beta_1 \cos \beta_2 - \cos \beta_1 \sin \beta_2)/2\end{aligned}$$

Since $x_1 = r_1 \cos \beta_1$, $x_2 = r_2 \cos \beta_2$, $y_1 = r_1 \sin \beta_1$, and $y_2 = r_2 \sin \beta_2$,

$$\begin{aligned}\text{Area} &= (r_1 r_2)((y_2/r_1)(x_2/r_2) - (x_1/r_1)(y_2/r_2))/2 \\ &= (x_2 y_1 - x_1 y_2)/2\end{aligned}$$

The area of each of these triangles is $(x_2 y_1 - x_1 y_2)/2$. If $(\lceil r/\sqrt{2} \rceil + 1)^2 + (\lfloor r/\sqrt{2} \rfloor)^2 > r^2$, there is a lattice point on the polygon where $y=x$, and adding these triangles creates a summation as follows:

$$\text{Area} = 8 \sum_{x=\lceil r/\sqrt{2} \rceil}^{\lfloor r/\sqrt{2} \rfloor} \left(x \lfloor \sqrt{r^2 - (x-1)^2} \rfloor - (x-1) \lfloor \sqrt{r^2 - x^2} \rfloor \right).$$

The number of boundary points on the polygon is $8(\lfloor r \rfloor - \lceil r/\sqrt{2} \rceil + 1)$, from above, so by Pick's Theorem, the number of interior lattice points for the polygon is $A - B/2 + 1$. However, to that we must add the number of points which are interior to the circle but on the boundary of the polygon, which is $B - B(r)$, so the total expression for the number of interior lattice points, $I(r)$, is $A - B(r) + 1 + B/2$, or:

$$I(r) = 8 \sum_{x=\lceil r/\sqrt{2} \rceil}^{\lfloor r/\sqrt{2} \rfloor} \left(x \lfloor \sqrt{r^2 - (x-1)^2} \rfloor - (x-1) \lfloor \sqrt{r^2 - x^2} \rfloor \right) - B(r) + 1 + 4(\lfloor r \rfloor - \lceil r/\sqrt{2} \rceil + 1)$$

If $(\lceil r/\sqrt{2} \rceil + 1)^2 + (\lfloor r/\sqrt{2} \rfloor)^2 \leq r^2$, there is no lattice point on the polygon where $y=x$, and an additional triangle must be added surrounding the lines $y=\pm x$. This triangle, by Pick's Theorem, has an area of $\lfloor r/\sqrt{2} \rfloor - 1/2$, so the total area of the polygon is:

$$\text{Area} = 8 \sum_{x=\lfloor r/\sqrt{2} \rfloor}^{\lfloor r \rfloor - 1} \left(x \lfloor \sqrt{r^2 - (x-1)^2} \rfloor - (x-1) \lfloor \sqrt{r^2 - x^2} \rfloor \right) + 4 \left(\lfloor r/\sqrt{2} \rfloor - 1/2 \right)$$

Proceeding on a similar path as above, we find:

$$I(r) = 8 \sum_{x=\lfloor r/\sqrt{2} \rfloor}^{\lfloor r \rfloor - 1} \left(x \lfloor \sqrt{r^2 - (x-1)^2} \rfloor - (x-1) \lfloor \sqrt{r^2 - x^2} \rfloor \right) + 4 \left(\lfloor r/\sqrt{2} \rfloor - 1/2 \right) - B(r) + 1 + 4 \left(\lfloor r \rfloor - \lceil r/\sqrt{2} \rceil + 1 \right)$$

These expressions lead to the same result as the far simpler expression found above by counting the points in an organized fashion:

$$I(r) = 4 \left(\sum_{x=1}^{\lfloor r \rfloor - 1} \lfloor \sqrt{r^2 - x^2} \rfloor \right) + 4 \lfloor r \rfloor - 3$$

We have expressions for $A(r)$, $B(r)$, and $I(r)$, that can all be evaluated exactly for a given r . These are:

$$I(r) = 4 \left(\sum_{x=1}^{\lfloor r \rfloor - 1} \lfloor \sqrt{r^2 - x^2} \rfloor \right) + 4 \lfloor r \rfloor - 3$$

$$B(r) = 8 \left(\prod_{j=1(4)}^{\lfloor r \rfloor} (1 + \alpha_j) \right) - 4$$

$$A(r) = \pi r^2$$

$A(r)$, since it is trivially solvable for r , can be related to either of the other two formulae by substituting for r . However, as neither $I(r)$ nor $B(r)$ are continuous, they cannot be written in terms of each other. Nonetheless, these expressions are in a way superior to Pick's theorem, as the values $I(r)$, $B(r)$, and $A(r)$ can be found from only one quantity, the radius. In Pick's theorem, knowledge of two of these quantities is required to calculate the third. All of the formulae work for all real values of the radius, so there are no special circumstances under which they are not valid.

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Euclidean Rings and an Application

BY LEONIDAS NGUYEN

It is often said that abstract theories are of little use in tackling concrete problems, but often in mathematics abstraction is a powerful tool that has a tremendously vast range of applications. For example, abstract ring theory results can be applied to prove such a result as *Any prime number 1 more than a multiple of 4 is expressible as the sum of two squares*. Indeed, it is the abstractedness of ring theory that makes it so useful.

A **ring** is a set with two binary operations, usually addition and multiplication, defined on it. A ring must satisfy certain properties:

- 1) Closure, commutativity, and associativity under addition.
- 2) Closure and associativity under multiplication.
- 3) Multiplication distributes over addition.
- 4) An additive identity and additive inverses for every element.

Additional convenient properties may or may not be satisfied, depending on the particular ring:

- 5) A multiplicative identity (unit element).
- 6) Commutativity under multiplication.

A ring with property 5) is called, naturally, a ring with unit element. A ring with property 6) is called commutative.

There are countless mathematical entities with ring structure. The set of all integers forms a ring with unit element; the set of all even integers forms a ring (without unit element). The set of all integers mod 6 is also a ring.

We are most familiar with number systems, so it is generally preferred that mathematical structures possess properties inherent in ordinary number arithmetic. The ring is not very ideal in this sense; for example, in the set of integers mod 6 we have $2 \cdot 3 = 0$. We are accustomed to the property $a \cdot b = 0$ implies at least one of a and b is zero. Mathematicians make the ring structure more "ideal" by introducing the concept of an **integral domain**, which is a commutative ring where the aforementioned property always holds true; i.e., there are no zero-divisors. (There are ways to further "idealize" the ring structure, but there is no need to consider that here.)

A special type of integral domain is a **Euclidean ring**. Such a ring (call it R) must satisfy the following properties:

- 1) We assign a nonnegative integer $d(a)$ to every nonzero element a in R .
- 2) If a and b are nonzero elements of R , $d(a)$ is less than or equal to $d(ab)$.
- 3) If a and b are nonzero elements of R , there are elements in R , call them q and r , such that $a = qb + r$ where either $r = 0$ or $d(r) < d(b)$.

Those familiar with the Euclidean algorithm should be able to see why such a ring has the name "Euclidean." Given a and b , we can divide a by b to obtain a quotient q and a remainder r "smaller" than b (we use the d -function to determine order or magnitude). For example, in the ring of integers, the function $d(a) = |a|$ satisfies the above properties. Another example of a Euclidean ring is the set of Gaussian integers $\mathbb{Z}[i]$, which are complex numbers in the form $a + bi$, where a and b are real integers. This property of the Gaussian integers can be used to prove an interesting theorem about prime numbers, mentioned at the beginning of this article. But first, some preliminary facts will be proven. The division properties of Euclidean rings can be extended to the prime factorization theorems that correspond very closely to ordinary arithmetic theorems

(for example, if a prime number p divides $a*b$, p divides at least one of a and b). This fact will be used in the ensuing proofs.

Lemma 1: *Let p be a prime integer and suppose that for some integer c relatively prime to p we can find integers x and y such that $x^2 + y^2 = cp$. Then p can be written as the sum of squares of two integers.*

It will be shown that p can be expressed as the product of two nonunit¹ Gaussian integers. First assume that this is not the case. Since $cp = x^2 + y^2 = (x+yi)(x-yi)$, p must divide one of $x+yi$ and $x-yi$ since $\mathbb{Z}[i]$ is Euclidean. Say p divides $x+yi$; then $x+yi = p(s+ti)$, which implies that $p(s-ti) = x-yi$, and so p divides both $x+yi$ and $x-yi$ and p^2 divides cp , contradicting the assertion that p and c are relatively prime. As a result we can express p as the product of two nonunits: $p = (a+bi)(c+di)$ (and also, as some computation will reveal, $p = (a-bi)(c-di)$). So $p^2 = (a^2 + b^2)(c^2 + d^2)$. Since $a + bi$ and $c + di$ are both nonunits, $a^2 + b^2 > 1$ and $c^2 + d^2 > 1$. The only factors of p^2 are 1, p , and p^2 , which implies $a^2 + b^2 = c^2 + d^2 = p$, and the result follows.

Lemma 2: *If p is a prime number of the form $4n + 1$, then there exists x such that x^2 is congruent to $-1 \pmod{p}$.*

Let $x = (1)(2)(3)\dots((p-1)/2)$. Since $p = 4n + 1$, $(p-1)/2$ is even and:
 $x = (-1)(-2)(-3)\dots(-(p-1)/2)$ (there are an even number of negative signs that cancel each other).
 $x^2 = (1)(2)\dots((p-1)/2)(-1)(-2)\dots(-(p-1)/2)$.

In mod p , however, -1 is the same as $p-1$, -2 is the same as $p-2$, and $-(p-1)/2 = (p+1)/2$, so
 $x^2 = (1)(2)\dots((p-1)/2)((p+1)/2)\dots(p-2)(p-1) = (p-1)!$ But $(p-1)!$ has a remainder of -1 upon division by p (Wilson's Theorem), and so x^2 is congruent to $-1 \pmod{p}$.

Theorem: *If p is a prime number that leaves a remainder of 1 upon division by 4, then p can be expressed as the sum of two squares.*

We use the result of Lemma 2, and take the remainder of x when it is divided by p (this can be done since we are working in mod p). The range of possible values for x can be restricted even further by taking the smaller of $p-x$ and x (the squares of each of these two numbers are congruent mod p). As a result we can pick an $x < p/2$ such that $x^2 + 1$ is a multiple of p (cp). Then $cp = x^2 + 1 < (p/2)^2 + 1 < p^2$, implying that c is relatively prime to p . Lemma 1 now applies, and we can conclude that p can be expressed as the sum of the squares of two integers.

¹ A unit is an element with an inverse. For example, -1 is a unit in the integers, since $-1 * -1 = 1$. The only units in the Gaussian integers are $1, -1, i, -i$.

Variations on the Sum of Two Squares Problem

BY SARA COHEN

The sum of two squares problem was first solved by Diophantus (325-409 A.D.) who posed the question: For what integers n will the equation $x^2 + y^2 = n$ yield integral solutions for x and y ?

Fermat provided the answer to this ancient problem, and characteristically offered no proofs of his conclusion. A generalization of Fermat's conjectures states that any integer of the form $2^a \prod_{p \equiv 1(4)} p^{\beta} \prod_{q \equiv 3(4)} q^{2\gamma}$, where p and q are primes, can be expressed as the sum of two squares.

This statement was later proven by Euler. Thus, this topic is quite old, but there are many interesting variations on the problem. For instance, consider the "two-squares like" equation:

$$(a^2 + 1)(b^2 + 1) = c^2$$

When is the product of a square plus one and another square plus one equal to a square? The trivial solution, when a equals b , is fairly obvious, but it turns out that there is a general equation for finding an infinite number of solutions. Surprisingly, this method involves Pell's Equation, which asks, what are the integral solutions to $x^2 - Ny^2 = \pm 1$, where N is a positive integer? Thus, before we attempt to solve the two-squares problem, two theorems on solutions to Pell's Equation are necessary.

Theorem 1: If (x_1, y_1) is a solution to $x^2 - Ny^2 = -1$, then all nontrivial solutions of the equation with x and y positive are given by $x_k + y_k \sqrt{N} = (x_1 + y_1 \sqrt{N})^k$ where k is odd, and solutions to $x^2 - Ny^2 = 1$, then all nontrivial solutions are given by $x_k + y_k \sqrt{N} = (x_1 + y_1 \sqrt{N})^k$ where k is even.

Theorem 2: If (x_0, y_0) is a solution to $x^2 - Ny^2 = d$, then all nontrivial solutions of the equation with x and y positive are given by $x_k + y_k \sqrt{N} = (x_1 + y_1 \sqrt{N})^k (x_0 + y_0 \sqrt{N})$ where (x_1, y_1) is a solution to $x^2 - Ny^2 = 1$ and $k = 1, 2, 3, \dots$

These theorems can be proven by induction without difficulty.

Rearranging the terms of $(a^2 + 1)(b^2 + 1) = c^2$ yields the equation: $c^2 - (b^2 + 1)a^2 = b^2 + 1$. This is, of course, Pell's Equation, and the solution $c = b^2 + 1$ and $a = b$ can be obtained by inspection.

Now we consider the equation $c^2 - (b^2 + 1)a^2 = -1$. The solution $c = b$ and $a = 1$ can be obtained by inspection, and by theorem 1, another set of solutions to the equation is given by $c^2 - (b^2 + 1)a^2 = 1$, which generates $c = 2b^2 + 1$ and $a = 2b$. By theorem 2, the final solution is:

$$c_k + a_k \sqrt{b^2 + 1} = \left[(2b^2 + 1) + 2b\sqrt{b^2 + 1} \right]^k \left[(b^2 + 1) + b\sqrt{b^2 + 1} \right]$$

For example, when $b = 1$:

$$c_1 + a_1 \sqrt{2} = [3 + 2\sqrt{2}]^1 [2 + \sqrt{2}] = 10 + 7\sqrt{2}, \text{ so } c = 10 \text{ and } a = 7$$

We can confirm that $(7^2 + 1)(1^2 + 1) = 10^2$ is indeed correct.

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Cardinality

BY LINDA HONG

Cardinality involves the study of "counting" infinite sets, such as the set of natural numbers and the set of rational numbers. But how can one count an infinite number of elements in such a set? Well, you don't really count, but you can use set theory to compare different types of sets and their cardinalities, or sizes.

A **mapping** or **map** is an association from each element in one set to exactly one element in another set. For example, take the sets S and S' and the mapping function f , which we denote as $f: S \rightarrow S'$. If x is an element of S , then $f(x)$ is an element of S' , and $f(x)$ is the **value** of f at x , or the **image** of x under f . Similarly, the set of all $f(x)$, or $f(S)$, a subset of S' , is the image of f and the image of S .

For the sets $S = \{a, b, c\}$ and $S' = \{r, s, t, u\}$ and the mapping $f: S \rightarrow S'$, we can define f so that $f(a) = r$, $f(b) = s$, and $f(c) = t$. Another way to denote this mapping is:

$$\begin{pmatrix} a & b & c \\ r & s & t \end{pmatrix}$$

Other possible mappings could be:

$$\begin{pmatrix} a & b & c \\ t & r & s \end{pmatrix} \text{ or } \begin{pmatrix} a & b & c \\ u & r & t \end{pmatrix} \text{ or } \begin{pmatrix} a & b & c \\ s & u & s \end{pmatrix}$$

In the first mapping, the a 's image would be r and the S 's image would be the set $\{r, s, t\}$, but not S' .

Some special mappings should also be considered here. $f: S \rightarrow S'$ is **injective** or **one-to-one** (1-1) if whenever $x, y \in S$ and $f(x) = f(y)$ then $x = y$. (This is like a regular one-to-one function where $f(x)$ is related to only one x .) The map is **surjective** if the image of f for $f(S)$ is equal to all of S' . In other words, for every $x' \in S'$ there exists an $x \in S$ such that $f(x) = x'$. Another way to put it is that f is **onto** S' . A mapping is said to be **bijective** or a **one-to-one correspondence** if it is both injective and surjective.

Let us take the set of all real numbers, $\mathbb{R}$. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be the mapping $f(x) = x + 1$. Now for every $y \in \mathbb{R}$ there exists a $(y-1) \in \mathbb{R}$. Therefore, f is a surjective or onto mapping, and since it can also be shown to be injective, f is a one-to-one correspondence. Another example of a bijection is the mapping $f: \mathbb{R}^+ \rightarrow \mathbb{R}$, where $f(x) = -x$. Here it is easy to see that for every positive real there exists a unique negative real, and vice-versa. This 1-1 correspondence also means that the two sets have the same number of elements.

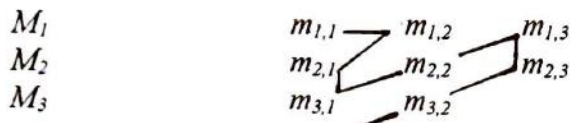
Now that we have defined the 1-1 correspondence, we can sufficiently compare sets to each other to determine its cardinality. For sets A and B , we say that the cardinality of A is the same as the cardinality of B , denoted as $\text{card}(A) = \text{card}(B)$, if there exists a 1-1 correspondence between A and B . $\text{card}(A) \leq \text{card}(B)$ if there exists an injective mapping $f: A \rightarrow B$.

A infinite set, S , is said to be **countable** if $\text{card}(S) = \text{card}(\mathbb{N})$, where $\mathbb{N}$ is the set of all natural numbers. Another way of saying this is that there exists a 1-1 function $f: S \rightarrow \mathbb{N}$. This means that S has the same number of elements as $\mathbb{N}$.

Let us visit the Ri Leebahd Fude Restaurant where unfortunately there are an infinite number of rats, and an infinite number of rat traps. The set of rats is $M = \{m_1, m_2, m_3, \dots\}$ and the

set of traps is $T = \{t_1, t_2, t_3, \dots\}$. Since both sets have the same cardinality as $\mathbb{N}$, $\text{card}(M) = \text{card}(T)$. However, another rat, m_0 , comes along. Will Mr. Fude have enough traps for all of them? Instead of mapping m_i to t_i , we can map each m_i to t_{i+1} . So there is still a one-to-one correspondence, and Mr. Fude can save some money.

Will another infinite set of rats need another infinite set of traps? No. If we add a new set of rats called $L = \{l_1, l_2, l_3, \dots\}$, then we can map each l_i to t_{2i} and each m_i to t_{2i-1} . The same thing will happen with an infinite number of infinite sets of rats. If we simply follow the algorithmic diagram shown below and assign each element a natural number, which corresponds to a rat trap, we will find that Mr. Fude is in luck that we were around when the invasion came on.



Using this method we can also find that the set of positive rational numbers is countable. If we let the first set be the set of all rational numbers with 1 as the numerator (and the natural numbers as denominators), the second set be the set with 2 as numerator, and so on, this proof would also hold. However, this would not work for the set of rats labeled with the real numbers.

Theorem 1: The set of all real numbers is not countable.

The following proof assumes the **Schroeder-Bernstein Theorem** which states: Given two sets S and T , if there exist a 1-1 map $f: S \rightarrow T$ and a 1-1 map $g: T \rightarrow S$, then there is a 1-1 correspondence between the two sets. This takes a little intuition to understand.

Proof: (By contradiction). First let us assume that $\mathbb{R}$ has the same cardinality as $\mathbb{N}$. We can then list the numbers as follows in decimal expansion:

$$\begin{aligned} r_1: & a_{10} . a_{11} a_{12} a_{13} \dots \\ r_2: & a_{20} . a_{21} a_{22} a_{23} \dots \\ r_3: & a_{30} . a_{31} a_{32} a_{33} \dots \end{aligned}$$

Each a_{i0} is the whole number part before the decimal point, and each a_{ij} , $j \neq 0$, is a digit after the decimal point. The r 's are assigned natural numbers. However, we can form a new number, x , as follows: we let b_1 be a digit not equal to a_{11} , and b_2 be a digit not equal to a_{22} , and so on as we go along the diagonal. In general, b_i is a digit not equal to a_{ii} . We then let $x = 0.b_1b_2b_3\dots$ so that x is not equal to any of the real numbers listed above. Since x cannot be mapped to any natural number, but is still a real number, $\mathbb{R}$ does not have a 1-1 correspondence with $\mathbb{N}$. Therefore the assumption is false, and $\mathbb{R}$ is not countable. (Note: $\text{card}(\mathbb{N}) < \text{card}(\mathbb{R})$ since there still is a mapping of $\mathbb{N} \rightarrow \mathbb{R}$).

The proof given above is known as Georg Cantor's famous **diagonal proof**. This proof distinctly shows that although $\mathbb{N}$ and $\mathbb{R}$ are both infinite sets, they still don't have the same number of elements. $\mathbb{R}$ is called **uncountable** for this reason.

Taking cardinality a step further, we can explore the cardinality of $\mathbb{R}$. An easy exercise for the reader is to prove that the cardinality of $(0, 1)$ is the same as the cardinality of $[0, 1]$. (Hint: Most proofs about cardinality involve the Schroeder-Bernstein Theorem). An even more interesting proof though, is the one that shows that $(0, 1)$ has the same cardinality as $\mathbb{R}$.

Proof: Since the definition of equal cardinalities say that a bijection has to be satisfied, it will suffice to show that there is a 1-1 function from $(0, 1)$ to $\mathbb{R}$. One such function is that can do this

is the tangent function. We can let $f(x) = \tan(\pi x - \pi/2)$. We can see that this proof is valid by graphing it.

Another way to prove more graphically is to take the line segment on the real number line from 0 to 1 and bend it into a semicircle. If we align the semicircle with the real number line, we see that a ray from the center of the circle and passing through a point on the circle will intersect the real line at one point (e.g., shining a flashlight through a transparent bowl).

Another useful theorem about cardinality has an ingenious but simple proof.

Theorem 2: $\text{card}(\mathbb{R}) = \text{card}(\mathbb{R}^2)$, $\mathbb{R}^2 = \{(a, b) : a, b \in \mathbb{R}\}$

Proof: Since it is easy to show that $\mathbb{R}$ maps onto $\mathbb{R}^2$, we will show that $\mathbb{R}^2$ maps onto $\mathbb{R}$.

Let the decimal expansion of (a, b) be $(\dots a_{-2} a_{-1} \cdot a_0 a_1 a_2 \dots, \dots b_{-2} b_{-1} \cdot b_0 b_1 b_2 \dots)$. We can define the function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ as $f(a, b) = \dots a_{-1} b_{-1} \cdot a_0 b_0 a_1 b_1 \dots$. This is a real number. Q.E.D.

In general, if A is a non-empty set, then $\text{card}(A) = \text{card}(A^2)$. (Since the proof of this is complicated I will not put it into this paper, but the reader can refer to Serge Lang's book Undergraduate Algebra for it). Assuming this is true, of course it naturally follows that $\text{card}(A) = \text{card}(A^n)$, with a simple proof by induction.

Another elegant theorem states that the set of all subsets of A , where A is infinite, has a greater cardinality than A .

Theorem 3: F , the set of all functions f such that $f: \mathbb{R} \rightarrow \mathbb{R}$, has a greater cardinality than $\mathbb{R}$.

Proof: For every f in F , there is a set of ordered pairs of real numbers that correspond, since a function is just a set of ordered pairs. f is therefore a subset of $\mathbb{R}^2$. Therefore F is just the set of all subsets of $\mathbb{R}^2$. $\text{Card}(F) > \text{card}(\mathbb{R}^2) = \text{card}(\mathbb{R})$.

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Metric Spaces and Applications

BY PING-HSIN LEE

By definition, a metric space is a function from ordered pairs of the elements of a set to the non-negative reals. More simply put, it is a set X with a distance function called a metric. It has several properties:

1. $d(x, y) = 0$ iff $x = y$
2. $d(x, y) = d(y, x)$ for $x, y \in X$
3. $d(x, y) + d(y, z) \geq d(x, z)$ for each $x, y, z \in X$

where d is a metric of distance on X , and X together with d is called the metric space (X, d) .

There are many examples of metric spaces, such as $\mathbb{R}^2$. More generally, $\mathbb{R}^n$ and the complex plane are also metric spaces. Discrete spaces (X, d) such as $d(x, y) = 0$ if $x = y$ and 1 if $x \neq y$ are also metric spaces.

Now, we can apply our knowledge of metric spaces to the field of sequences. By definition, a sequence is simply an infinite list; one example of a sequence is $1, 2, 3, 4, \dots$. We will focus on one property of sequences: convergence. In a metric space (X, d) the sequence $x_1, x_2, \dots, x_n, \dots$ of points of X converges to limit x , $x \in X$, if $d(x_n, x) \rightarrow 0$ (or if the sequence $d(x_1, x), d(x_2, x), \dots$ of real numbers converges to 0) as $n \rightarrow \infty$. We can simplify the notation by writing $x_1, x_2, \dots \rightarrow x$ and $x_n \rightarrow x$ as $n \rightarrow \infty$.

Theorem 1: If $x_1, x_2, \dots$ is a convergent sequence in metric space (X, d) , then $d(x_m, x_n) \rightarrow 0$ as $m, n \rightarrow \infty$.

Proof: Since $x_1, x_2, \dots$ is a convergent sequence, we can let x be the limit of that sequence. Then we have $0 \leq d(x_m, x_n) \leq d(x_m, x) + d(x, x_n)$. Since $d(x_m, x) + d(x, x_n) \rightarrow 0$ as $m, n \rightarrow \infty$, $d(x_m, x_n) \rightarrow 0$ as $m, n \rightarrow \infty$. Q.E.D.

Sequences for which $d(x_m, x_n) \rightarrow 0$ as $m, n \rightarrow \infty$ are often referred to as Cauchy sequences. A sequence $x_1, x_2, \dots$ is bounded above if there exists a number k such that $x_n \leq k$ for all n . Similarly, a sequence is bounded below if there exists a number j such that $j \leq x_n$ for all n . Therefore, we can say that a sequence is bounded if it lies entirely within the upper and lower bounds. A common assumption in real analysis is that a sequence which is increasing and bounded must be convergent. From this we can easily prove that a bounded sequence has a convergent subsequence. We shall proceed assuming this theorem without proof. This theorem will be applied later in complete sets.

Another field of mathematics that we can apply metric spaces to is set theory. By definition, in a metric space (X, d) the set $A \subseteq X$ is closed if whenever $a_1, a_2, \dots \in A$ and $a_1, a_2, \dots \rightarrow a$ it follows that $a \in A$. Now we can prove two important theorems concerning closed sets.

Theorem 2: Any intersection of closed sets in a metric space is itself closed.

Proof: Let B be the intersection of any number of closed sets in a metric space and let $b_1, b_2, \dots$ be points of B with $b_1, b_2, \dots \rightarrow b$. We wish to show that $b \in B$ to prove that B is closed. But since $b_1, b_2, \dots$ are in the intersection of all the closed sets in question, they are contained in each of these

closed sets. So the limit b is also in each of the closed sets. Therefore we can say that b is in the intersection of all these closed sets, or $b \in B$. So B is closed. Q.E.D.

Theorem 3: Let $B_1, B_2, \dots, B_k$ be closed sets in a metric space. Then $B_1 \cup B_2 \cup \dots \cup B_k$ is closed.

Proof: Let $B = B_1 \cup B_2 \cup \dots \cup B_k$ and let $b_1, b_2, \dots, b_k \in B$ with $b_1, b_2, \dots \rightarrow b$. We wish to show that $b \in B$ to prove B is closed. But we know that some B_i must contain an infinite number of $b_1, b_2, \dots$ because otherwise $B_1, B_2, \dots, B_k$ would only contain in total a finite number of $b_1, b_2, \dots$. Therefore we can assume that the subsequence $b_{k_1}, b_{k_2}, \dots$ lies in B_i . Also we know that $b_{k_1}, b_{k_2}, \dots \rightarrow b$ and so $b \in B_i$ since B_i is closed. Therefore $b \in B_1 \cup B_2 \cup \dots \cup B_k = B$ and B is closed. Q.E.D.

Metric spaces can also be applied to complete sets. By definition, the set $A \subseteq X$ is complete in the metric space (X, d) if whenever $a_1, a_2, \dots$ is a sequence in A with $d(a_m, a_n) \rightarrow 0$ as $m, n \rightarrow \infty$, it follows that the sequence is convergent with its limit in A . There are many examples of complete metric spaces (also called complete spaces) such as $\mathbb{R}$, $\mathbb{R}^2$, and $\mathbb{R}^n$.

Theorem 4: If $A \subseteq B$ in a metric space and A is closed and B is complete, then A itself is complete.

Proof: Let $a_1, a_2, \dots$ be in A (therefore also in B) with the property that $d(a_m, a_n) \rightarrow 0$ as $m, n \rightarrow \infty$. Then we can say that by the completeness of B , $a_1, a_2, \dots \rightarrow b \in B$. But we know that $a_1, a_2, \dots$ are all contained in the closed set A , and therefore $b \in A$. So A is complete. Q.E.D.

Theorem 5: Let A be a complete set in a metric space. Then A is closed.

Proof: Consider the contrapositive: If A is not closed, then A is not complete. Since A is not closed, we can let a convergent sequence $a_1, a_2, \dots$ with its limit $a \notin A$. Then by theorem 4, $d(a_m, a_n) \rightarrow 0$ as $m, n \rightarrow \infty$. But $a_1, a_2, \dots$ does not converge to a point of A , therefore A is not complete. Q.E.D.

The final section of set theory deals with compact sets. By definition a set A in a metric space (X, d) is compact if every sequence in A has a subsequence convergent to a point of A . We know that from the definition any closed and bounded intervals $[a, b]$ in $\mathbb{R}$ are therefore compact. However, $\mathbb{R}$ itself is not compact. This is because the sequence $1, 2, 3, 4, \dots$ has no convergent subsequence.

Theorem 6: Let A be a compact set in a metric space. Then A is complete.

Proof: Let A be compact in the metric space (X, d) and let $a_1, a_2, \dots$ be a sequence in A with the property that $d(a_m, a_n) \rightarrow 0$ as $m, n \rightarrow \infty$. We wish to show that the sequence $a_1, a_2, \dots$ converges to a point of A . Since A is compact, there exists a convergent subsequence $a_{k_1}, a_{k_2}, a_{k_3}, \dots \rightarrow a$ with its limit in A . Also, we know that $0 \leq d(a_m, a) \leq d(a_m, a_{k_n}) + d(a_{k_n}, a)$ is true. But $d(a_m, a_{k_n}) \rightarrow 0$ since $(a_m, a_n) \rightarrow 0$ as $m, n \rightarrow \infty$ and $d(a_{k_n}, a) \rightarrow 0$ since $a_{k_1}, a_{k_2}, a_{k_3}, \dots \rightarrow a$. Therefore, $d(a_m, a) \rightarrow 0$ or $a_1, a_2, \dots \rightarrow a \in A$. Therefore, A is complete. Q.E.D.

Now we can see the relationship between compact, complete and closed sets. We can see that by theorems 5 and 6 that the following holds true: **compact** $\Rightarrow$ **complete** $\Rightarrow$ **closed**. This shows that these three different kinds of sets are progressively "stronger."

Lastly, metric spaces can be used with the contraction mapping principle. By definition, a contraction of the metric space (X, d) is a function $f: X \rightarrow X$ with the property that, for some real number $k < 1$, $d(f(x), f(y)) \leq kd(x, y)$ for all $x, y \in X$. Also, a fixed point of f occurs when $x = f(x)$.

Theorem 7: Let $f: X \rightarrow X$ be a contraction of the complete metric space (X, d) . Then f has a unique fixed point. Furthermore, if x_1 is any point of X , then the sequence $x_1, x_2 = f(x_1), x_3 = f(x_2), \dots$ converges to that unique fixed point.

Proof: Let x_1 be any point of X and let $x_2 = f(x_1), x_3 = f(x_2), \dots$. We wish to show that the sequence $x_1, x_2, x_3, \dots$ converges and that its limit is the unique fixed point of f . First, we want to see if $x_1, x_2, x_3, \dots$ is a Cauchy sequence. But before calculating the distance from x_n to x_m , consider the distance from x_n to x_{n+1} . From the definition of contractions and the given of the theorem we have the following inequalities:

$$\begin{aligned} d(x_n, x_{n+1}) &= d(f(x_{n-1}), f(x_n)) \leq kd(x_{n-1}, x_n) \\ &= kd(f(x_{n-2}), f(x_{n-1})) \leq k^2 d(x_{n-2}, x_{n-1}) \\ &\vdots \\ &= k^{n-2} d(f(x_1), f(x_2)) \leq k^{n-1} d(x_1, x_2) \end{aligned}$$

We can see that $d(x_n, x_{n+1}) \leq k^{n-1} d(x_1, x_2)$. So, for large n , the distance between x_n and x_{n+1} is very small since $0 \leq k < 1$. Now we can use this inequality to find $d(x_n, x_m)$ for $m > n$. We have

$$\begin{aligned} d(x_n, x_m) &\leq d(x_n, x_{n+1}) + d(x_{n+1}, x_m) \\ &\leq d(x_n, x_{n+1}) + d(x_{n+1}, x_{n+2}) + d(x_{n+2}, x_m) \\ &\vdots \\ &\leq d(x_n, x_{n+1}) + d(x_{n+1}, x_{n+2}) + d(x_{n+2}, x_{n+3}) + \dots + d(x_{m-1}, x_m). \end{aligned}$$

Since we have already established an upper bound for each of the successive x 's, we have

$$\begin{aligned} d(x_n, x_m) &\leq d(x_n, x_{n+1}) + d(x_{n+1}, x_{n+2}) + \dots + d(x_{m-1}, x_m) \\ &\leq k^{n-1} d(x_1, x_2) + k^n d(x_1, x_2) + \dots + k^{m-2} d(x_1, x_2) \\ &\leq (k^{n-1} + k^n + \dots) d(x_1, x_2) \\ &= k^{n-1} (1 + k + k^2 + \dots) d(x_1, x_2) \\ &= k^{n-1} d(x_1, x_2) / (1 - k) \end{aligned}$$

But $k^{n-1} d(x_1, x_2) / (1 - k) \rightarrow 0$ as $n \rightarrow \infty$, which means that $d(x_n, x_m) \rightarrow 0$ as $n \rightarrow \infty$. Therefore $x_1, x_2, x_3, \dots$ is a Cauchy sequence in the complete metric space (X, d) . So $x_1, x_2, x_3, \dots \rightarrow x$.

Then $f(x_1), f(x_2), f(x_3), \dots \rightarrow f(x)$ since the distance between $f(x_n)$ and $f(x)$ is less than that of x_n from x , or $x_2, x_3, \dots \rightarrow f(x)$. But we know that also $x_2, x_3, \dots \rightarrow x$, therefore $x = f(x)$. So we've proven that the limit of the sequence is indeed a fixed point of f .

For the second part of the proof that it is the unique fixed point of f , we let (X, d) be a metric space and let $f: X \rightarrow X$ have the property that $d(f(x), f(y)) < d(x, y)$ for all $x \neq y$ in X . We wish to show that if $f(a_1) = a_1$ and $f(a_2) = a_2$, then $a_1 = a_2$, or that f has at most one fixed point.

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Affine Planes

BY TZYU-MING YEH

Euclidean geometry, the most familiar form of geometry, is based upon Euclid's five postulates. Perhaps the most famous one is the *Parallel Postulate*, which states that: through a point not on a line, there exist one and only one line parallel to the original line. From these basic postulates, which must be assumed to be true for Euclidean geometry, many of the theorems and common properties can be derived.

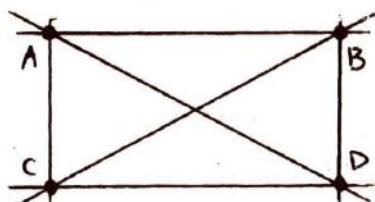
Like Euclidean geometry, other fields of geometry can also be formed from a set of axioms or postulates. One such field is affine planes. The three main axioms that affine planes are based on are as follows:

1. Given two distinct points P , Q in the set of points, there exists one and only one line (l) with the property that both P and Q are on the line. A counter-example is the following diagram:

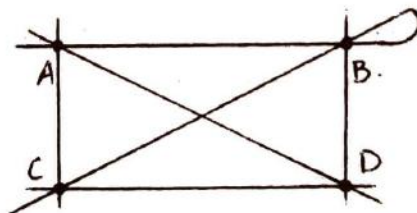


This figure cannot exist in an affine plane because there are two line passing through points P and Q . This violates the first axiom;

2. If l is a line and P is a point not on l , there exists one and only one line l_1 such that P is on l_1 and l and l_1 are parallel;
3. There exists four points on the plane such that no three of these are contained in a common line. The only figure that these four points can have is the following:



A counter-example of this third axiom is the following diagram which is a plane including only four points:



The points A , B , and C are all on the same line, and since there are only four points on this affine plane, it is a violation of the third axiom.

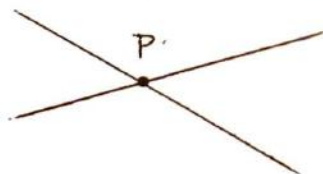
A Comparison of the Real Plane and Affine Plane on an Elementary Level:

The following is a line segment PQ :



On the ordinary (Euclidean) plane, there exist an infinite number of lines between points P and Q . On the affine plane, however, when considering this line segment, there exist only two points, P and Q . This also holds true in the comparison of a real plane and an affine plane. There are an infinite number of points in a real plane, but the number of points on an affine plane is just the number of labeled points that are drawn.

The next diagram is a set of crossing lines:



For the ordinary plane, there always exists a point through which both lines cross. This is the point of intersection. If this diagram were in an affine plane, there does not necessarily have to be a point of intersection. Unless, this "point" is given a name, there is no point that exists where these two lines cross. One can think of two lines that appear to cross from a bird's eye view, but are actually at different heights.

A line is a very important concept in the real plane – it is the shortest distance between two points. On the other hand, there is no concept of length in the affine plane, and a line does not have to be the shortest distance between the two points. A line simply connects two points that exist on the plane.

Two of the axioms for the affine plane is the same for an ordinary plane. The first axiom: given two distinct points, there exists one and only one line through both points, is also true for the real plane. The second axiom: given a line and a point not on the line, there exists one and only one line parallel to the original line (this is the Parallel Postulate in Euclidean geometry). Additionally, if there exist two lines on either an affine plane or real plane, the two lines intersect at no points or exactly one point.

Some Theorems of the Affine Plane

Theorem 1: Through a point there has to exist at least three lines that pass through that point (assuming that there are at least four points in the plane, because if there are less than three points then there do not have to exist three lines).

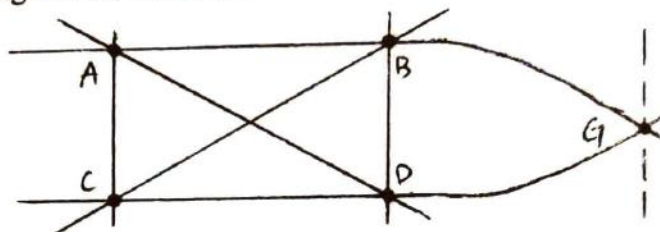
The proof can be divided into two cases; the first case is where the point is one of the special four points that are formed from axiom three and the second case is where the point is not one of the four special points. By inspecting the figure of the special four points:



it is already evident that through each of the four points, there already exist three lines. If an extra point is added, from this new point to each of the old points there will be a line. Therefore, if the point is one of the special four, there exist at least three lines that pass through the point.

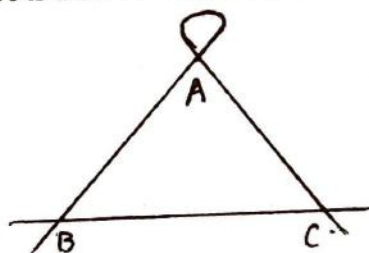
In the second case, the point in consideration is not one of the special four points. For an affine plane with more than four points, there still must exist the special four figure due to axiom three. For any added point G , there has to be lines that are drawn from these special four points. Extending these lines is the best way to show that this theorem is true because it minimizes the

number of lines that have to pass through the new point. Observing all possibilities in which lines can be extended from the six existing lines, there is one possibility. There can be only two lines that are drawn from the six lines. If more of the lines are used, there will be a violation of the first axiom. The new figure now looks like:



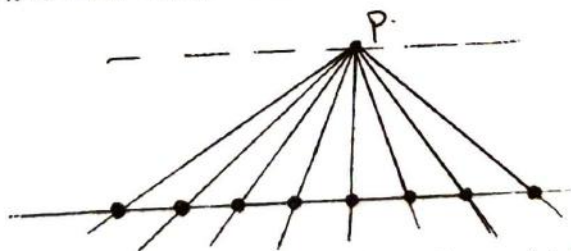
From this, two lines already pass through G . There has to exist a third line that passes through this point in order to satisfy the second axiom. As a result, through the point G , there are three lines.

Theorem 2: Through three distinct points A, B, C , of the three lines AB, AC, BC , either all three lines are different or all the lines are the same. One way to prove this theorem is to first assume that $AB = AC$, then show that there is a contradiction if $BC \neq AB$. From the diagram that can be drawn from these assumptions:

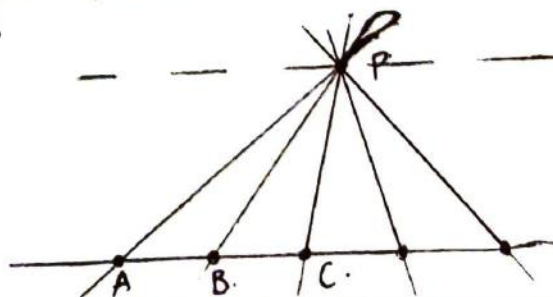


it can be seen that through the points B and C , there exist two different lines that pass through both of them. This is a contradiction because it violates the first axiom. Therefore if $BC \neq AB$ is true, then $AB = AC$ cannot be true because of the contradiction. If $AB = AC$, then all three lines have to be the same to avoid the violation of axiom one.

Theorem 3: Let P be a point and l be a line not containing P . If l has n points, then P is contained in exactly $n+1$ lines. There is line that is drawn from point P to each of the points on line l .



Since there are n points, there are n lines that are drawn from point P to line l . All of these lines have to be distinct (which gives the n lines) because if they were not, then there is a violation of axiom one. From the next diagram,

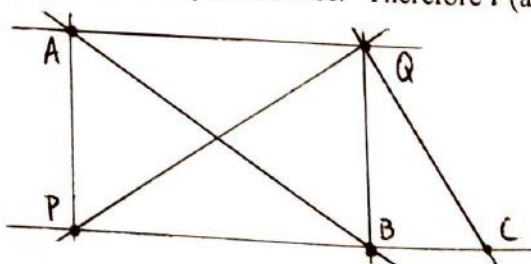


it can be seen that both APB and ABC pass through points A and B if the lines AP and PB are the same. This is a violation of axiom one. The lines are therefore all distinct and there are n lines in total. Applying axiom two, through point P , there exists a line that is parallel to line l . This line is not one of the first n lines because it will never cross line l at a point. There are therefore a total of $n+1$ lines.

Theorem 4: In the affine plane, either all lines have the same number of points or there are 2 parallel lines that contain all the points in the affine plane. The first part of the theorem can be proven by using the converse of theorem 3. Assume there are two lines that intersect, l and l_1 . There has to exist a point P so that a parallel line to l and l_1 may be drawn. If there are n points on line l , then to point P there are $n+1$ lines. From the $n+1$ lines that are drawn through P , there will be n lines that intersect l_1 in n points.



Therefore l and l_1 have the same number of points. Applying this a few more times, all the lines in the affine plane will have the same number of points. The second half of the theorem involves two parallel lines. If the lines were not parallel, there would have to be a point not on either line that provides the new line that is parallel to both l and l_1 . This theorem holds true for the special four points from axiom three. Next, assume that one of the lines has at least three points. There are then two lines that can be drawn BQ and CQ , but both of them are parallel to AP . There is a contradiction and the assumption is false. Therefore l (and similarly for l_1), must have exactly two points.



The latter part of the theorem is true only for a four point affine plane, the special four points from axiom three.

The Order of An Affine Plane

The order of affine plane is the number of points on each line. For an affine plane, if the order is n , then there are n^2 points in total. This can be shown by first considering a certain point K . All the lines will pass through this point, and therefore every point is collinear to K . There are $n+1$ lines that are in consideration. To avoid repeating the point that is being used for reference, there are $n-1$ points per line. Multiplying, there are n^2-1 points excluding K . There are n^2 points in all if the order of an affine plane is equal to n .

Secondly, if there are n points per line (order of n), then there are a total of n^2+n lines that exist in the affine plane. There are two ways to show that this is true. First, there exist n^2 points in an affine plane of order n . All possible line segments that can be obtained from this is given by ${}_nC_2$ (C stands for a combination), where a line segment is determined by two points. Some of

these line segments are actually on the same line. To find out the number of possible line segments on the same line, there are ${}_nC_2$. To find the number of distinct lines, divide the first combination by the second, or: ${}_nC_2 \div {}_nC_2 = \frac{(n^2)(n-1)}{n} \div \frac{n(n-1)}{2} = n(n+1)$. There are $n^2 + n$ distinct lines in an affine plane of order n . The second way to do this is to realize that the number of points is given by: (# lines)(# points/line)/(# lines/point). There are a total of n^2 points in the plane. The number of points per line is just n (by definition, the order of the affine plane) and there are $n+1$ lines that pass through each point. Putting all this into the equation, $n^2 = \frac{n(\# \text{ lines})}{n+1} \Rightarrow \# \text{ lines} = \frac{n^2(n+1)}{n} = n^2 + n$. Similarly, by this method, there are still $n^2 + n$ lines that exist in an affine plane of order n .

Equivalence Classes of Affine Planes

Let S be a set and if this relation $\sim$ satisfies the following three properties, it is called an **equivalence class**.

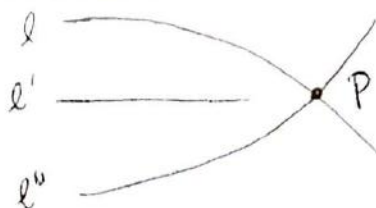
1. Symmetry Property: ($a \sim b \Rightarrow b \sim a$)
2. Transitive Property: ($a \sim b \wedge b \sim c \Rightarrow a \sim c$)
3. Reflexive Property: ($a \sim a$ is true)

An example of an equivalence relation is equality, "=", in the set of real numbers. These properties are known to be true in the set of reals: if $a = b$ then $b = a$, if $a = b$ and $b = c$ then $a = c$, and $a = a$ is always true. The relation ">" in the set of reals fails because the first and third properties are violated. It is true that $a > b$ or $b > a$, but never both. Furthermore, by definition, a cannot be greater than a .

A basic theorem of equivalence relations is: given a set S , the equivalence class generated by a is denoted as $E(a)$, or the set of all elements b such that $a \sim b$. The following is always true, either $E(a) \cap E(b) = \emptyset$ or $E(a) = E(b)$. The equivalence classes break the set into totally disjoint pieces and there are no overlaps.

Consider the relation parallel for the affine plane. By the basic theorem of equivalence relations, it can be shown that the relation parallel in the ordinary plane is an equivalence relation. This can be done by considering the slopes of lines in the plane. $E(l_0)$ is the class of lines that have the same slope. Each line has a unique slope and the lines can be placed into an equivalence class. There will be disjoint sets because the lines can only have one slope. Since the equivalence classes completely separate the set of lines by slope, parallel in the ordinary plane is an equivalence relation. The basic theorem of equivalence cannot be used for parallel lines in an affine plane, so the initial approach must be taken.

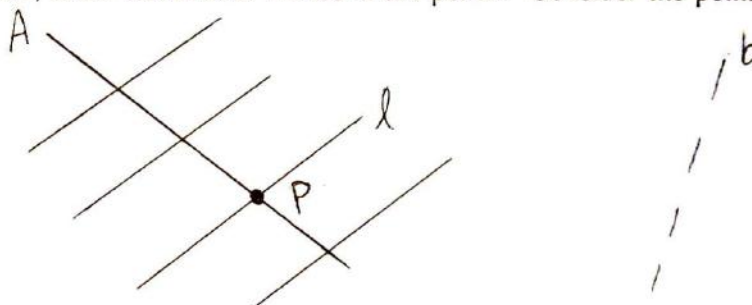
By definition, two lines in an affine plane are parallel if $l = l'$ or l and l' do not intersect. Denote the relation parallel as $\sim$. We need to show that l is always $\sim l$, if $l \sim l'$ then $l' \sim l$, and if $l \sim l'$ and $l' \sim l''$ then $l \sim l''$. The first property by definition is always true because a line is parallel to itself. Secondly, if l and l' do not intersect at any point, then l' and l do not intersect in any point or in other words $l' \sim l$ given $l \sim l'$. Also if $l = l'$, then $l' = l$. This still means that the lines are parallel. For the third property, there are four cases that are possible, but three of them are trivial. This triviality comes from the former part of the definition for parallel in an affine plane. If two or more of the lines are equal, then there is just a bunch of substitutions that can take place. The last case is where all three lines are distinct. Assume that $l \sim l''$ and that they intersect at P :



From the given information, l' cannot contain P because l' is parallel to both l and l'' . Therefore through the point P , there exists two lines that are parallel to l' . This violates the second axiom of an affine plane and there is a contradiction. Therefore, $l \sim l''$ from this proof by contradiction. All three properties are true, and parallel in an affine plane is an equivalence class.

The equivalence classes set by the relation parallel in an affine plane are completely disjoint. Each equivalence class of parallel lines is called a **pencil**. Consider the affine plane of order n . Every pencil of parallel lines contains exactly n lines. Furthermore, the total number of pencils in the affine plane is $n+1$.

By the second axiom, there are at least n lines in the pencil. Consider the point P on the diagram below:



Other than P , there are $n-1$ points. Each one of these points will give a unique line because there cannot be more than one parallel line per point that is parallel to l . There are $n-1$ lines that exist from these $n-1$ points. Line l is the n th line. There does not exist anymore lines. Assume there was a line b , but it cannot intersect A or l anywhere. This line cannot exist because both A and l would be parallel to b through the point P . This is a contradiction of the second axiom. There cannot exist another line. There are n lines total in a pencil of an affine plane of order n .

The number of pencils per affine plane uses the fact that there are n lines in each pencil.

The following equality holds true: $\# \text{ lines} = \left(\frac{\# \text{ lines}}{\text{pencil}} \right) (\# \text{ pencils})$. There are n^2+n lines in an affine plane. Furthermore, there are n lines per pencil. Therefore, $n^2+n = n(\# \text{ pencils})$. From this, there are $n+1$ pencils in an affine plane of order n and n lines per pencil.

Proof that $\mathbb{R}^2$ (Real Plane) Forms an Affine Plane:

$\mathbb{R}^2$ can be written as a set of ordered pairs: $\{(a,b) \mid a,b \in \mathbb{R}\}$ and a collection of subsets which are really the lines. The collection of lines has different slopes and it may be separated into those lines that have a nonzero slope and those lines with the zero slopes. For the former, let $l_{m,b} = \{(x, mx+b) \mid x \in \mathbb{R}\}$ for every choice of m and b in $\mathbb{R}$. For the latter, let $l^c = \{(c,y) \mid y \in \mathbb{R}\}$ for every choice of c in $\mathbb{R}$. The theorem that has to be proved now is that $\mathbb{R}^2$ is an affine plane.

The proof consists of following the three axioms and seeing if they all apply in $\mathbb{R}^2$. The first axiom is: given two distinct points (g,h) and (n,k) , there exist exactly one line that contains both points.

The first case is if $g = n$. No line of the form $l_{m,b}$ can pass through the two points because the x coordinate has to be different for each ordered pair. Consider the second case l^c for a line.

In this line, all the x coordinates are the same and therefore one and only one line $\ell^x (= \ell^n)$ will contain the two points.

The second case is if $g = n$. No line of the form ℓ^x can contain both points because the x coordinates are different for the two points. Therefore $l_{m,b}$ has to contain the two points. If it is to contain the two points, by substitution:

$$\begin{aligned} h &= mg + b \text{ and } k = mn + b && \text{subtracting the two terms} \\ (h - k) &= mg + b - (mn + b) = mg - mn = m(g - n) && \text{solving out for } m \\ m &= (h - k) / (g - n) = (h - k)(g - n)^{-1} && \text{by substituting into the original} \\ h &= (h - k)(g - n)^{-1}g + b && \text{solving out for } b \\ b &= h - (h - k)(g - n)^{-1}g. \end{aligned}$$

From this since the subtraction and multiplication of real numbers always gives an unique answer, m and b are unique for each given two points. The line that contains both points is therefore $l_{(h-k)(g-n)^{-1}, h-(h-k)(g-n)^{-1}g}$. This is the unique line that will contain both of the points. To check this, plug the points into the derived equation.

The second axiom is to check if the parallel postulate holds true. Given a point (p,q) and a line l not containing (p,q) , there has to exist an unique line such that l' contains (p,q) and l and l' have no points in common.

The problem can again be broken into two different cases, the first being where the line $l = l_{m,b}$. l' cannot be of the form ℓ^x because the two lines will intersect at $(c,md+b)$. l' can not be of the form $l_{n,d}$, where $n = d$ because both lines will contain the point $((d-b)(m-n)^{-1}, m(d-b)(m-n)^{-1}+b)$. By messy calculations, this can also be shown true. Therefore the problem is if there exists a line of the form $l_{m,d}$ that satisfies axiom two. The axiom has two parts to it: one is that it contains the point (p,q) and the other is that it will not intersect the original line of the form $l_{m,b}$. $l_{m,d}$ passes through (p,q) iff $q = mp + d$, or $d = q - mp$. Therefore there is only one unique line that will pass through (p,q) and d has to equal $(q - mp)$. If $l_{m,b}$ and $l_{m,d}$ were suppose to be distinct lines, $d \neq b$. Assume that the lines intersect at a point (f,g) . There are two equations, $g = mf + b$ and $g = mf + d$. From this, $b = g - mf = d$, which leads to a contradiction. The assumption is false and the two lines do not intersect and are distinct.

The second case is where the line is of the form, $l = \ell^x$. $p = c$, but l' has to equal ℓ^x . By definition of the line, (p,q) is in the line ℓ^x . It is the only line that passes through the point (p,q) that is of the form ℓ^x and it cannot be of the form $l_{m,b}$ because there would then be an intersection at $(c,mc+b)$. Therefore there is only one line that passes through (p,q) that is parallel to the original line l .

The final task is to show that the third axiom holds. Assume that there are two different elements, r and s , in $\mathfrak{R}$. The four points under consideration would be (r,r) , (r,s) , (s,r) , and (s,s) . If three of them were collinear, two must have the same x coordinate. The third point with the different x coordinate would not lie on this line because $r \neq s$. Therefore there is a contradiction and three of them cannot lie on a common line when there are four points.

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Special Thanks to: Allen Sheppard, a teacher, at Hampshire College *Studies in Mathematics* (1995).

The Königsberg Bridge Problem

BY VINCI LIU

One of the most fascinating problems in the history of mathematics is the Königsberg Bridge Problem. It appealed to the masses because one can *attempt* to solve the problem without any knowledge of mathematics. As some might expect, the problem originated from a place called Königsberg. The Pregel River runs through this town in Prussia, leaving an island in the middle and branches out in two. Seven bridges connect Königsberg, as shown in Figure-1 below. The Königsberg Bridge Problem is whether or not a person can walk across each of the seven bridges once and only once.

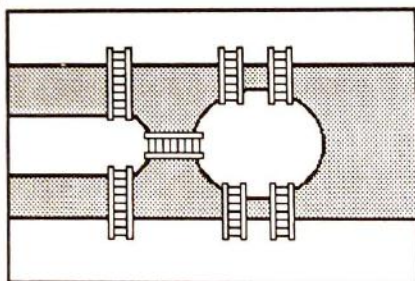


Figure - 1

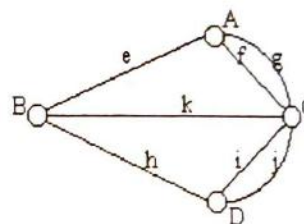


Figure - 2

While anyone can attempt to find a solution to the Königsberg Bridge Problem, it took a famous Swiss mathematician, Leonhard Euler, to solve. He proved that it was impossible for a person to cross the seven bridges of Königsberg once and only once. In addition, he generalized his solution as a theorem to solve the problem for *any* configuration of land and bridges.

The first step Euler took to solve the problem is to represent Figure-1 as a multigraph. A multigraph consists of vertices (shown as circles) and edges (lines) that join two vertices. Each land mass and each bridge in Königsberg is represented by a vertex and an edge, respectively. (See Figure-2 above).

The path sought by our person in Königsberg is called an Eulerian Trail. A trail is defined as a sequence of alternating vertices and edges, starting and ending on vertices, and having no repeated edges. One can find a trail by tracing a path from any vertex, without repeating an edge, to any other vertex. For example, {A, e, B, k, C, i, D, j, C} is a trail. An Eulerian trail further requires that each edge in the graph is included in the trail (which is the "each of the seven bridges once and only once" requirement in the Königsberg bridge problem).

Euler stated that no Eulerian trail existed in the graph of Figure-2 because it contained four vertices of odd degree. The degree of a vertex is the number of edges incident to it (number of edges ending at the vertex). But why does four vertices of odd degree preclude the possibility of an Eulerian trail? The explanation is that an Eulerian trail cannot pass through a vertex of an odd degree. An Eulerian trail must enter the vertex using one edge and leave using another edge. Since there are an odd number of edges, one edge will be left in the end which the Eulerian trail must pass through. Vertices of odd degree can only be the beginning or the end of an Eulerian trail. The graph of Figure-2 contains four vertices of odd degree. Thus, the Eulerian trail must pass through two of the vertices. As already demonstrated, this is impossible!

Euler's theorem for solving Königsberg Bridge Problem for any configuration is as follows:

*A graph contains an Eulerian trail if and only if it has zero or two vertices of odd degree.**

This theorem has two part. First, "If a graph contains an Eulerian trail, then it has zero or two vertices of odd degree." Second, "If a graph has zero or two vertices of odd degree, then it contains an Eulerian trail." The explanation above can be generalized to prove the contrapositive of the first part ("If a graph does not have zero or two vertices of odd degree, then it does not contain an Eulerian trail"), and thus the first part. The second part of Euler's theorem is harder to prove. The following is an informal proof.

For simplicity, assume that the graph G has two vertices of an odd degree. Use induction on the number of edges. A graph with one edge (and thus two odd vertices) contain an Eulerian trail. Assume that any graph with k or less edges contain an Eulerian trail. It suffices to show that any graph with $k+1$ edges contain an Eulerian trail.

Randomly construct a trail T from one of the vertex of odd degree to the other. If this contains all edges, then we are done. If not, then we remove the edges of this trail from the graph. The resulting graph G' contains all vertices of even degree (by removing T , an odd number of edges is removed from the starting and ending vertex and an even number of edges is removed from other vertices). The resulting graph G' has all vertices of even degree and k or less edges. By the assumption, it contains Eulerian circuits $C_1, C_2, \dots, C_n$. (If G' is "fragmented" by the removal of T , then multiple number of Eulerian circuits results). All the Eulerian circuits start and end on vertices passed by the original trail. Thus, one can construct a Eulerian trail E in graph G by joining the Eulerian circuits $C_1, C_2, \dots, C_n$ to T . Q.E.D.

* A graph with no vertex of odd degree contains an Eulerian circuit, a more specific form of Eulerian circuit. An Eulerian circuit starts and ends on the same vertex. A graph cannot have an odd number of vertices with odd degree.

Braid Theory

BY HYUN-SUP BYUN

Professor Sylvain Cappell of the Courant Institute of New York University gave a series of lectures at Stuyvesant High School on topology.¹ Professor Cappell, a renowned topologist, began with the concept of braids.

Braid theory, originated by Emil Artin during the 1920's, involves methods of weaving strings on an imaginary board. Consider k nails arranged on a line at the top of a board, and another k nails similarly arranged at the bottom. If one braids k strings on this board (with each nail holding only one string) and each string only descends (e.g., does not make a loop with itself), one has a braid. The set of all k -stranded braids is denoted as B_k . Figure 1 includes two examples of 3-stranded braids. Note that at an intersection, one string is "broken" to show that it passes under the other. Also note that the length of the strings is of little concern to us.

The next step up from the textile industry is codify braids and develop an algebraic system to represent them. A useful step is to employ sigma braids and tau braids. First, we number the "nails" at the top of the braid sequentially. If the only two strings that cross are the i th and $(i+1)$ st strings, and the i th string crosses over the $(i+1)$ st, then the braid is called σ_i (see Figure 2a). If the $(i+1)$ st string crosses over the i th string, then the braid is called τ_i (see Figure 2b).

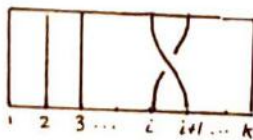


Figure 2a



Figure 2b

The next question is: What kind of operation on the two braids would retain both of their properties? We can define braid multiplication as attaching the ends of one braid to those of the other (see Figure 3). Clearly, the operation is closed, since the product of two k -stranded braids is another k -stranded braid. Second, the operation is associative – it doesn't matter in what order we multiply braids, so long as the sequence of braids is unchanged. Third, there is an identity element e : the k -stranded unknot, which has no intersections, only straight lines.

Fourth, do products of braids commute? Well, many do commute. Consider 4-stranded braids σ_1 and σ_3 . By inspection, the $\sigma_1\sigma_3$ is the same as $\sigma_3\sigma_1$: they don't interfere with each other.

So commutativity holds sometimes, and it holds for all 2-stranded braids. However, $\sigma_1\sigma_2$ is not equal to $\sigma_2\sigma_1$ (see Figure 4b), so commutativity does not hold in general.²

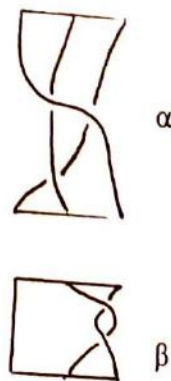


Figure 1

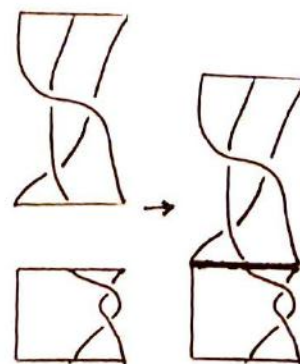


Figure 3

¹ There are still two more lectures left at the time of this writing.

Fifth, every k -stranded braid has an inverse, although this may not be obvious at first. If we multiply σ_i by τ_i , the product is the identity braid e . Thus, they are inverses of each other. Furthermore, we can decompose any braid as a product of σ_i 's and τ_i 's: simply stretch the braid out and separate out the intersections. This is possible because the strings only descend. Each intersection is merely a sigma or tau braid, and thus we can rewrite the original braid as a product of these. Every sigma or tau braid has an inverse, as noted earlier, so we can thus find the inverse of the original expression. Note that multiplication does not necessarily commute, so each "factor" must be arranged properly: for example, the inverse of $\sigma_1\tau_3\tau_5$ is $\sigma_5\sigma_3\tau_1$ and not $\tau_1\sigma_3\sigma_5$.

Students of algebra should immediately recognize the set of k -stranded braids and multiplication to be a group. Indeed, this system is called the Artin Braid Group.

B_2 , the set of all 2-stranded braids, is slightly unusual: it is an Abelian group. Furthermore, $B_2 = \{(\sigma_1)^m \mid m \in \mathbb{Z}\}$, so for every integer, there is a 2-stranded braid, and vice versa. Algebra students will also note that B_2 is isomorphic to $\mathbb{Z}$.

A logical successor to braid theory is knot theory. Knots are generally represented by closed loops with knots, and we project the knots onto a flat surface. In fact, a theorem of Markov's states that any knot can be generated by a corresponding braid; one merely runs another string "behind" the board to close the loops (see Figure 4). A major problem in knot theory, as in Artin's Braid Groups, is the classification of knots.

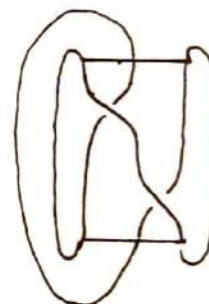


Figure 4

One method is to count the number of crossings in the knot. For example, the trefoil (Figure 5) is the only knot with three crossings. For four and five crossings, there is only one knot each (see Figure 6). However, the number of knots with a given number of crossings rises quite rapidly: fifty or so knots have nine crossings, while there are thousands for ten.

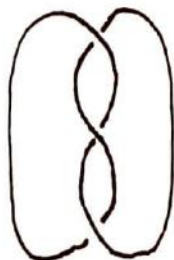


Figure 5

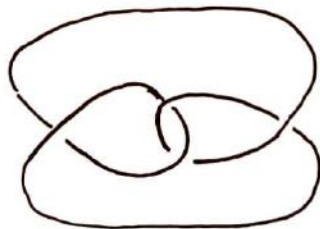
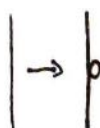
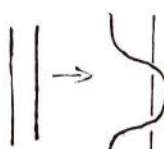
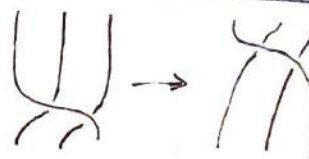


Figure 6

Aside from magnitude, there is another problem. How do we know if two knots are equivalent? Actually, there is really no algorithm to automatically compare two knots. One method is to try and deform a knot, without changing to make it identical to another.

Reidemeister's theorem states that: Two projections yield the same link or knot iff they can be obtained from each other by a Reidemeister move. There are three types of legal deformities, or Reidemeister moves. The first one is to twist the string to make a small loop with itself, and this is the move R_1 (see Figure 7a). The second is to pull one string across another to create a small loop; this is R_2 (see Figure 7b). The third is to pull a string up to a new position without changing the loop; this is R_3 (see Figure 7c).

² Interestingly, if we consider $\sigma_1\sigma_2\sigma_1$, it is in fact equal to $\sigma_2\sigma_1\sigma_2$. This is quite similar to the Yang-Baxter equation in physics, which solves for two matrices A and B such that $ABA = BAB$. Braid theory provides an interesting analogy, and similar techniques may apply to both fields.

Figure 7a R_1 Figure 7b R_2 Figure 7c R_3

A corollary to Reidemeister's theorem states: Suppose we have an invariant that is computed from a projection. If this invariant is unchanged by a Reidemeister move, then it only relies on the underlying knot. Thus the challenge becomes that of identifying some invariant.

One candidate is the linking number. In order to employ linking numbers, the loops must be oriented; i.e., we must assign directions that the loops will follow (see Figure 8). We define ϵ at a crossing to be equal to ± 1 , depending on the crossing (see Figure 9), and the linking number is equal to $\frac{1}{2} \sum \epsilon$. However, the linking number, at best, can only provide a negative response; even if two knots have the same linking number, there is no guarantee that they are equivalent.



Figure 8



$$\epsilon = +1$$



$$\epsilon = -1$$

Figure 9

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Chaotic Mathematics

BY NELSON UHAN

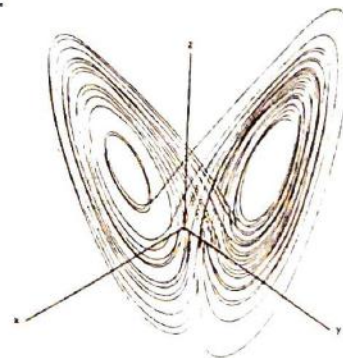
People often view mathematics as a very orderly and predictable subject. Yet, one area of math truly defies this view: chaos theory. What is chaos theory? No, it does not involve surrealism or anarchy, but rather the behavior of arbitrary nonlinear systems, or systems where the elements are not linked in a linear or proportional manner. To understand these types of systems, mathematicians often build models and simulations, employing mathematical formulas, actual physical models, and computer simulations.

In 1960, a meteorologist named Edward Lorenz, while playing with a computer model of a weather system, discovered that by plugging in numbers that varied ever so slightly from his original model of the weather, even differences as small one-thousandths would result in weather projections which soon bore no discernible relation to the original. This was the beginning of the mathematical science of chaos theory.

Although Lorenz's computer model was quite primitive, his inference was correct: the enormous divergence between the computer-simulated weather systems was caused by the tiny variations in the system's initial conditions. As mathematicians found out, generally, systems governed by nonlinear rules and equations can be extremely sensitive to minuscule changes, often showing unforeseeable and "chaotic" behavior as a result. In contrast, linear systems behave "normally," where small differences in initial conditions lead to only small differences in final outcomes.

The weather, even when presented in its simplest model, would not be amenable to long-range prediction because the system is very sensitive to minuscule changes in the initial conditions. Small variations snowball into large deviations, and the entire process occurs in a very nonrepetitive and unpredictable manner. Even though certain general constraints in the system are satisfied, (i.e. general seasonal temperature gradients), specific long-range predictions would be worthless.

Since Lorenz, there have been many manifestations of the so-called "Butterfly Effect" (the sensitive dependence of nonlinear systems on their initial conditions) in subjects ranging from biology (including heart fibrillations and epilepsy) to economics (price fluctuations) and to hydrodynamics (turbulence and fluid flow). These nonlinear systems demonstrate great complexity even with systems defined by the simplest rules and equations. Below is a diagram of the Lorenz attractor. It is shaped like a butterfly – hence the "Butterfly Effect." If we chart the trajectory of a system in chaotic motion in a higher dimensional mathematical "phase space" – where it is completely unpredictable, it is impossible to take a given point in time and predict where it will end up at a later point. Surprisingly, the motion of the chaotic system is not completely random, as one could clearly see from the diagram below.



However, effects of chaos may not be readily discernible. This is partly due to the fact that these systems often behave quite normally but suddenly become chaotic when a parameter of the system reaches a critical value. For example, take physicist Mitchell Feigenbaum's extremely simple model of the population of a certain animal species. The population is given by the nonlinear formula $x' = rx(1-x)$, where x' is the population one year, x is the population the preceding year, and r is a parameter that varies from 0 to 4. To add further simplicity, let's confine x and x' to be numbers between 0 and 1, the true population being 1,000,000 times the resulting values.

Let's assume that the population x is 0.3 (actually 300,000) and that r is 2. Plugging in the values, we obtain 0.41. The following year, the population would be 0.4838. Eventually, we find that the population stabilizes at 0.5:

x	x'
0.3	0.41
0.41	0.4838
0.4838	0.4994751
0.4994751	0.4999994
0.4999994	0.5
0.5	0.5
0.5	0.5
0.5	0.5
0.5	0.5

Similar calculations with a smaller r , say $r = 1$, demonstrate that whatever its initial value, the population stabilizes at 0, and it becomes extinct. When r is larger than 2, say 2.6, we find that the population now stabilizes at 0.62, no matter what the initial value for population was.

What is the point of all these calculations? Well, let's now assume that $r = 3.2$. We know that the initial value for population does not matter, but when we substitute into the formula $3.2x(1-x)$, we find that the population no longer stabilizes at one value, but eventually settles into an alternation between two values:

x	x'
0.21	0.53088
0.53088	0.796948562
0.796948562	0.517828965
0.517828965	0.79898281
0.79898281	0.513949695
0.513949695	0.799377299
0.799377299	0.513194345
0.513194345	0.79944291
0.79944291	0.53106862
0.53106862	0.799453476
0.799453476	0.513048371
0.513048371	0.799455168

One value stabilizes at about approaching 0.513, and the other at approximately 0.799. In other words, on year the population is 0.5, and the next 0.8. If we raise the parameter r to 3.5, we find that the population alternates regularly among four different values, approximately 0.38, 0.83, 0.50, 0.88, in successive years. If we again increase r ever so slightly, the population settles into a regular alternation among eight different values. Smaller and smaller increases in r lead to more doublings of the number of values alternating (If you don't believe me, you can try it for yourself.).

But when we reach $r = 3.57$, the population never stabilizes, and instead varies randomly from year to year (Try putting the equation into a spreadsheet and copy the equation all the way to the bottom of the sheet. You'll never find a pattern.). What would happen if we increased r again ever so slightly? The population would still vary randomly, right? Wrong. In fact, if we increase r again, it would result in a regular alternation of the species' population from year to year, and a further increase in r would lead again to the chaotic variation. Think about it: if a system as trivial as this single nonlinear equation can demonstrate such chaotic unpredictability, imagine how chaotic other systems would be, such as various social, economic, and ecological policies with numerous equations and rules involved.

One might conclude that these chaotic, nonlinear systems are rare, and would only interest the math nerd and various "weirdo scientists". But now that more tools for dealing with nonlinear have been developed, all types of scientists are realizing that it is no longer necessary to screen

out noise, static, and friction, or to ignore turbulence, and "random" contingency. New chaos-aware control techniques are being used to stabilize lasers, manipulate chemical reactions, encode information, and change chaotic heart rhythms into healthy, regular heart rhythms. Meanwhile, mathematics is becoming more unpredictable and disorderly than before.

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136	43	118	25	100	7	82
43	44	45	46	47	48	49
61	115	22	97	4	79	133
36	37	38	39	40	41	42
112	40	94	1	76	130	58
29	30	31	32	33	34	35
37	91	19	73	127	55	109
22	23	24	25	26	27	28
88	16	70	145	52	106	34
15	16	17	18	19	20	21
13	67	142	49	124	31	85
8	9	10	11	12	13	14
64	139	46	121	28	103	10
1	2	3	4	5	6	7

Answers to Magic Square on page 40:

A Summer at PROMYS

BY HYUN-SUP BYUN

As I recall, a group of my friends were taking refuge in my dorm room when my counselor walked in. He then brandished three rings in order to execute a hoary magic trick. After a few quick manipulations (his dexterity with stupid puzzles was unparalleled), yes, the rings were linked. "Now, how did I do it?" the Rubiksmaster dared.

"Rings are closed under addition," someone ventured.

Too much math during the summer? Probably not, but the spirit of mathematics tends to permeate one's mind at PROMYS. PROMYS¹, a summer mathematics program at Boston University, is geared towards this nefarious goal. High school students leave indoctrinated in quadratic reciprocity, $\mathbb{Z}[\sqrt{5}]$, twenty-four, and the number-color game.

The daily PROMYS diet of mathematics consisted only of a short morning lecture on number theory, problem sets, and review sessions. Lectures were short in order to encourage one to enjoy the problem sets during one's free time. The program intends that students learn more from working on problem sets than from class, via pondering, conjecturing, and proving – the foundations of mathematical research. For examples, lectures often provide guidance and leads over outright answers. In practice, it is much more satisfying to work out an elegant proof by oneself than by copying it from the board.

The basic curriculum is elementary number theory, ranging from bases to factorization domains. One of the big thrusts of the program is the law of quadratic reciprocity, which is a simple-looking law with a beautiful and non-obvious derivation. More advanced students can supplement number theory with another seminar. For example, PROMYS offered Spheres, Combinatorics, and Topology last year; this year, there will be Spheres, Class Numbers, and Hyperbolic Geometry. There are smaller classes ("counselor minis") for more advanced topics. This year, they are also encouraging individual research (e.g., for Westinghouse projects).

PROMYS attempts to ensure that students do not fall behind permanently. Every four students are assigned to an experienced counselor, who can work closely to provide help as well as correct the problem sets (which should be handed in daily). For example, one counselor with a heavy Texas accent used to work daily with a French student. During review sessions, held several times per week, a counselor or professor will go over any requested questions.

PROMYS is a medium-sized program, with approximately seventy students from around the U.S. (and last year, France). Everyone has a roommate, and friends often cooperatively attack the problem sets. The work isn't overly onerous, but it can be difficult and it is not unusual to see unusual reactions to math develop. When another BU summer program held a Jeopardy tournament, someone reportedly added categories such as "P-adic numbers" for them to enjoy.

To apply, contact:

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¹ i.e., the Program for Young Scientists in Mathematics. Yes, it's an irregular name, and no, it's not pronounced "pro-meece."

The Pit and the Pendulum—A Mathematical Spoof

By Tommy Wu

I was sick—sick unto death of math. For days, I had been a prisoner of the math teachers. They had tortured me by droning endless formulas and theorems to me. Finally, the sentence the dread sentence of death by mathematics reached my ears, coming from the white lips of the teachers. I would be given three puzzles to solve. If I solved them correctly, I would be freed. If not, I would be killed. Then silence, and stillness, and night were the universe.

I had swooned. When I came to, I was in pitch blackness. I heard a voice coming from an unseen source. It said:

“A traveler comes to a fork in the road and does not know which way to go to reach his destination. There are two men at the fork, one of whom always lies while the other always tells the truth. The traveler doesn’t know which is which. He may ask one of the men only one question to find his way. What is his question and which man does he ask?”

I got up and began to pace the room. This was a tricky one. At length, though, I came up with the answer. I said out loud, “Ask either man, ‘If I were to ask you if this is the way I should go, would you say yes?’ If the man he asks is the truth-teller, he will of course get the right answer. If the man he asks is the man who always lies, that man lies about the answer he would give, thus giving the correct answer.”

As I said this, the lights suddenly came on. I found myself at the edge of a deep pit, in the middle of the room. So, this was their plan! They asked me a tough question so that I would pace the floor thinking about it. If I hadn’t gotten the answer, I would have fallen to oblivion.

Shaking, I stumbled back to the wall. I resolved to stay awake, but I fell asleep anyway. When I woke, I found a pitcher of water beside me. I drank it and grew drowsy again. It must have been drugged. I slipped into unconsciousness.

When I opened my eyes, I found that I was bound by leather straps. I looked up and saw a picture of Albert Einstein on the ceiling. In his hand, he held a huge metal pendulum. As I looked more closely, I saw that the pendulum was swinging. After a while, I saw that the heavy razor was actually descending. I had escaped the pit, so now the fiends were going to slice me in half with the pendulum.

A voice boomed: “If some coffee is 97% caffeine free, how many cups of it would one have to drink to get the amount of caffeine in a cup of regular coffee?”

I began to sweat. I looked up and saw that the mighty blade had swung considerably lower.

Down—steadily down it crept. I cursed those darn cups of coffee!

Down—certainly, relentlessly, down! It vibrated within three inches (or should I say 7.62 centimeters, by the metric system?) of my bosom!

Down—still unceasingly—still inevitably down! I gasped and struggled at each vibration. I hated my math teachers with all my heart!

Suddenly, it came to me. How could I be so stupid?! “ $33\frac{1}{3}$ cups. Because there is 3% caffeine left in the doctored coffee; in 100 cups there would be enough for 3 cups of regular; 3 goes into 100 exactly $33\frac{1}{3}$ times.”

As soon as I said this, the pendulum drew back up and my bonds were mysteriously loosened. I was free.

Free!—and in the grasp of the Inquisition! I looked at the walls, where the pictures of demonic mathematicians were outlined. Pascal, Archimedes, Euclid, Fermat—the whole crew was there. I noticed that the figures were beginning to glow.

Unreal!—even while I breathed there came to my nostrils the breath of the vapor of heated iron. The walls began to close in on me. I backed towards the center of the room, where the pit was, and I heard:

“A hunter wakes up at his camp one morning. He walks south for half a mile and trips on a rock. Then he walks south another half a mile. He sees a bear and chases it. He runs east for one mile and then gives up. So he walks north one mile back to his camp. Unfortunately, a second bear had torn up the camp. What color is the bear that tore up the camp?”

What?! What kind of a question is that?! I couldn't even begin to find an answer. The closing walls pressed me onward. As I tottered upon the brink, I screamed in agony and despair—

The fiery walls rushed back! There was a blast of trumpets! An outstretched arm caught my own as I fell, fainting, into the abyss. It was that of General Quinlan. The English Teachers' Army had entered New York. The Mathematical Inquisition was in the hands of its enemies.

Answer to Final Problem

White. The only place where someone can go south one mile, east one mile, and north one mile, and end up at the same place is the north pole.



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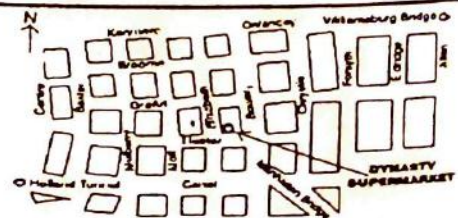
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Wordsearch

BY BRANDON IRRIZARY

- The property of being even or odd is called _____.
- The sentence "This sentence is false" is known as an _____, named after its dictator.
- The sentence in [2] is an example of a _____ paradox.
- A man by the name of _____ began his monumental article on AI with the sentence "Can machines think?"
- The *pons asinorum* proof involves a/an _____ triangle.
- The governmental system we know of as _____, coming from the Greek words for "people" and "rule, power" from the _____ has been accused of being mathematically unsound.
- Legend has it that _____, enraged, composed a problem about grazing cattle.
- This problem represents a branch of mathematics known as _____.
- This mathematician and philosopher called the _____ numbers "a wonderful flight of God's spirit."
- Concerning the distribution of bridge suits, the most probable one is _____ [the numbers of each suit translate into letters.]
- The phrase "Eadem mutata resurgo" refers to this type of spiral _____.
- The word "digital" refers to a segmented, divided concept. The word for an unsegmented, continuous process is _____.
- With the sole exception of $2/3$, for which there was a special hieroglyph, the Egyptians had symbols only for fractions with this number as the numerator _____.
- The three-dimensional version of an unfolded six-sided playing die is known as an unfolded _____.
- This is known as the "Shoemaker's knife" of Archimedes: _____.
- Charles Sanders Peirce was fond of triads;
The Greeks, however, were fond of _____.
- π multiplied by i^4 equals _____.
- The base of the Sumerian number system was _____.
- Of all fields of mathematics, the Greeks excelled at this field: _____.

L	S	I	X	T	Y	E	I	G	H	T	Y	O	N	E	Z	E	R	P	S
I	A	W	O	D	S	C	P	E	E	P	A	M	T	N	T	S	E	E	E
K	R	R	E	P	U	B	L	I	C	A	N	I	S	M	I	T	A	R	D
L	C	H	I	O	X	D	N	H	Y	R	S	F	T	Y	N	G	O	T	I
I	H	A	N	P	I	D	Y	A	T	B	E	I	E	M	B	Y	O	I	M
K	I	V	A	O	S	O	E	Y	N	E	L	E	T	O	E	E	N	T	E
L	M	E	H	S	E	C	T	I	E	L	E	N	R	O	I	R	E	S	N
I	E	A	A	A	M	N	I	R	C	O	C	D	A	Y	L	R	E	P	I
K	D	H	H	K	A	O	N	M	C	S	S	C	D	R	D	Y	E	N	P
H	E	E	A	R	N	N	O	S	H	A	O	U	P	O	E	N	H	O	E
A	S	O	T	V	T	A	N	P	S	T	S	L	I	F	I	I	D	E	U
H	A	N	L	U	I	O	M	E	N	S	I	Y	T	I	R	A	P	M	O
E	N	E	L	A	C	S	O	R	R	Y	S	R	A	S	F	L	E	T	S
E	I	I	E	G	N	I	R	U	T	N	A	L	A	A	T	O	P	J	U
Y	N	O	N	Y	R	T	E	M	O	E	G	A	L	G	T	V	S	I	D
N	D	R	I	G	H	T	T	H	I	S	W	A	Y	E	O	E	O	N	H
H	Y	P	E	R	C	U	B	E	E	P	L	E	A	S	G	L	G	A	E
A	A	B	I	L	G	I	S	I	Z	L	I	K	L	E	R	I	A	I	H
N	D	S	E	V	G	I	Y	R	O	E	H	T	R	E	B	M	U	N	R
D	E	M	O	C	R	A	C	Y	A	U	S	T	R	A	L	I	A	F	A

15. number theory
16. Gottfried Leibnitz
17. 4, 4, 3, 2
18. logarithmic spiral
19. analog
one
hypercube

1. parity
2. Epimenides
3. semantic
4. Alan Turing
5. isosceles
6. democracy
7. Archimedes

MAGIC SQUARE by Che King Leo

1.	2.	3.	4.	5.	6.	7.
8.	9.	10.	11.	12.	13.	14.
15.	16.	17.	18.	19.	20.	21.
22.	23.	24.	25.	26.	27.	28.
29.	30.	31.	32.	33.	34.	35.
36.	37.	38.	39.	40.	41.	42.
43.	44.	45.	46.	47.	48.	49.

1. can be expressed as a cube or a square
2. difference between the 69th and 70th squares
3. How tall in inches is Bart Simpson if he is 3 and 5/6 ft tall?
4. $5^n - 4$
5. sum of the first seven natural numbers
6. second number not accounted by #46
7. number of ways to choose 2 of 5 ellipses
8. an unlucky number
9. sum of itself and its opposite is 143
10. one less than the product of twin primes
11. a perfect square whose digits are also perfect squares
12. #47 plus a factorial
13. $6^n - 5$
14. area of a rhombus with diagonals 10 and 17
15. number of keys on an old piano
16. $n^x = x^n$
17. only four distinct pairs of primes whose sums are x
18. one more than a 3 digit number with property of #11
19. number of cards in a poker deck
20. #47 plus a factorial
21. a Fibonacci number
22. prime opposite #25
23. product of two primes differing by six
24. $58979 / 90720 + 47023 / 134400 = a / b$; a, b relatively prime. Find b-a.
25. prime opposite #22
26. $2^n - 1$
27. Fibonacci number following #21
28. the sum of any x consecutive integers is always divisible by 109
29. $x_{\text{base } 3} = 14_{\text{base } 10}$
30. radius of circle with equation $x^2 + y^2 - 16x + 12y - 1500 = 0$
31. geometric mean of 16, 188 and 2209
32. fits the description of #4, #13, #26, #36, #41, #44
33. ninth number in the Lucas series
34. $\#47 + {}_6P_2$
35. approximate area of an ellipse with axes length 8 and 9.232
36. $4^n - 3$
37. $5^8 \bmod 330$
38. approximation for the Prime Number Theorem for 100
39. last number counted for #46
40. $n^n = n + n = x$
41. $3^n - 2$
42. a Fibonacci number minus #47
43. an even multiple of 17
44. $7^n - 6$
45. greatest common divisor of 2006 and 3894
46. number of primes less than 100
47. $\log_{10}(\log_{10}(\text{googolplex}))$
48. $(1/73)$ of the magic sum
49. $n^4 + m^4$; where m, n are odd

Answers on page 35.

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The staff of *Math Survey* deeply appreciates and thanks you for your support.

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